

Quartic Elimination under Epistemic Sovereignty Compression

A Theoretical Framework for Polynomial Closure in AI-Geopolitical Systems

Soumadeep Ghosh

Kolkata, India

Abstract

This paper develops a formal theoretical extension of epistemic sovereignty theory under advanced artificial intelligence, computational concentration, geopolitical competition, and informational warfare. Building upon the Triple-AI/ML framework and the Six-Nation Bound, we introduce the Quartic Elimination Theorem: if three nations possess total epistemic sovereignty, then a fourth nation can force the remaining geopolitical system into a quartic sovereignty constraint whose equilibria determine the survivability of all residual sovereign actors.

The framework integrates mathematics, economics, finance, geopolitics, warfare economics, cybernetics, AI/ML systems theory, graph theory, network economics, international relations, algorithmic concentration, and computational statecraft into a unified analytical structure. We demonstrate that increasing returns in compute infrastructure, semiconductor concentration, data centralization, and financial-information leverage compress the global sovereignty manifold into algebraically constrained forms.

The paper formalizes sovereignty compression as a recursive reduction of geopolitical degrees of freedom. Under strong concentration dynamics, the remaining sovereignty manifold collapses into quartic closure relations. The resulting polynomial system determines whether independent informational sovereignty can survive under advanced AI conditions.

The framework suggests that sovereignty in the AI age becomes fundamentally computational, recursive, and algebraically constrained.

The paper ends with “The End”

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1 Introduction

The emergence of advanced artificial intelligence transforms the structure of geopolitical power. Historically, sovereignty depended primarily upon:

1. territorial control,
2. industrial capacity,
3. military force,
4. fiscal extraction,
5. population scale.

The AI age introduces new strategic primitives:

1. frontier model generation,
2. semiconductor dominance,
3. compute sovereignty,
4. data centralization,
5. narrative engineering,
6. cybernetic governance,
7. financial-information leverage,
8. military-AI integration.

As these systems scale, informational sovereignty becomes increasingly concentrated. The resulting geopolitical order exhibits structural compression driven by increasing returns in computation and informational infrastructure.

This paper develops a mathematical and geopolitical framework for understanding quartic sovereignty elimination under advanced AI concentration dynamics.

2 Epistemic Sovereignty

Define epistemic sovereignty as the ability of a nation-state to autonomously:

1. generate strategic information,
2. control computational infrastructure,
3. produce frontier AI systems,
4. sustain military AI operations,
5. control informational narratives,
6. maintain financial-information autonomy.

Let the sovereignty vector of nation i be:

$$\mathbf{S}_i = (A_i, C_i, D_i, N_i, M_i, F_i)$$

where:

$$\begin{aligned} A_i &= \text{Frontier AI capability} \\ C_i &= \text{Compute sovereignty} \\ D_i &= \text{Data sovereignty} \\ N_i &= \text{Narrative sovereignty} \\ M_i &= \text{Military-AI integration} \\ F_i &= \text{Financial-information leverage} \end{aligned}$$

Assume normalization:

$$0 \leq S_{ij} \leq 1$$

for all dimensions.

Define total epistemic sovereignty by:

$$\|\mathbf{S}_i\|_1 = 6$$

where:

$$\|\mathbf{S}_i\|_1 = \sum_{j=1}^6 S_{ij}$$

3 The Six-Nation Bound

Let:

$$N_E$$

denote the number of fully epistemically sovereign nations.

The Six-Nation Bound conjectures:

$$N_E \leq 6$$

under advanced AI concentration dynamics.

The central mechanism is increasing returns.

3.1 Compute Concentration

Let effective compute stock be:

$$K_i$$

for nation i .

Compute sovereignty is:

$$C_i = \frac{K_i^\gamma}{\sum_j K_j^\gamma}$$

where:

$$\gamma > 1$$

captures increasing returns.

As γ rises:

1. compute centralizes,
2. frontier AI centralizes,
3. semiconductor leverage increases,
4. sovereignty compresses.

4 Triple Concentration Dynamics

Suppose three nations:

$$X_1, X_2, X_3$$

already possess total epistemic sovereignty.

These nations dominate:

1. semiconductor fabrication,
2. GPU production,
3. hyperscale compute,
4. frontier AI models,
5. reserve currency systems,
6. satellite infrastructure,
7. strategic data accumulation,
8. cybernetic warfare systems.

Define aggregate sovereignty concentration:

$$\Omega_3 = \sum_{i=1}^3 \omega_i$$

where ω_i denotes normalized sovereignty share.

Residual sovereignty capacity becomes:

$$R = 6 - \Omega_3$$

As concentration intensifies:

$$R \rightarrow 0$$

Thus the remaining geopolitical system loses independent sovereignty dimensionality.

5 Residual Sovereignty Manifold

The remaining nations no longer possess independent six-dimensional sovereignty spaces. Instead, their feasible sovereignty region collapses onto a lower-dimensional manifold. Let:

$$x$$

denote residual sovereignty amplitude for all remaining nations collectively. The system exhibits recursive interactions across:

1. compute recursion,
2. data recursion,
3. financial recursion,
4. military recursion.

These four recursive strategic loops generate quartic closure.

6 The Quartic Sovereignty Equation

The residual sovereignty equilibrium satisfies:

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

where:

- a = Compute recursion coefficient
- b = Data concentration coefficient
- c = Narrative-financial coupling coefficient
- d = Military-AI dependence coefficient
- e = Structural sovereignty offset

The quartic determines all admissible sovereignty equilibria for residual states.

6.1 Interpretation

Positive real roots correspond to:

1. survivable sovereignty states,
2. stable informational autonomy,
3. computational independence.

Complex or negative roots correspond to:

1. informational dependence,
2. sovereignty collapse,
3. algorithmic reducibility.

7 Quartic Elimination Theorem

7.1 Statement

Theorem 1. *Suppose:*

1. *three nations possess total epistemic sovereignty,*
2. *AI systems exhibit increasing returns,*
3. *compute infrastructure is capacity constrained,*
4. *semiconductor production is oligopolistic,*
5. *informational systems exhibit recursive coupling.*

Then there exists a fourth sufficiently capable nation X_4 such that the remaining sovereignty dynamics reduce to a quartic polynomial constraint:

$$P_4(x) = 0$$

whose roots determine admissible sovereignty equilibria.

Moreover, strategic forcing conditions may be selected such that no stable positive real roots exist.

Hence all remaining nations become informationally dependent.

7.2 Proof Sketch

Suppose:

1. *compute scales superlinearly,*
2. *semiconductor access is geographically concentrated,*
3. *reserve currency leverage centralizes globally,*
4. *narrative systems exhibit network effects,*
5. *military AI depends upon compute access.*

Then:

1. *dominant nations accumulate disproportionate strategic advantages,*
2. *smaller states lose independent compute access,*
3. *AI training becomes externally dependent,*
4. *informational autonomy collapses recursively.*

Residual sovereignty dynamics become recursively constrained through four dominant strategic loops:

1. *compute recursion,*
2. *data recursion,*
3. *finance recursion,*
4. *military recursion.*

Thus the feasible sovereignty manifold reduces to quartic closure:

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

By appropriate strategic forcing, coefficients may be selected such that:

1. *all positive real roots vanish,*
2. *all equilibria become unstable,*
3. *sovereignty becomes reducible.*

Therefore residual nations become informationally dependent.

□

8 Examples of Strategic Quartics

8.1 No Real Sovereignty Equilibrium

Consider:

$$x^4 + 4x^2 + 16 = 0$$

No real roots exist.

Interpretation:

1. no independent sovereignty equilibrium survives,
2. all remaining states become structurally dependent.

8.2 Degenerate Controlled Equilibrium

Consider:

$$(x - 1)^4 = 0$$

The system admits a single degenerate equilibrium.

Interpretation:

1. sovereignty exists only in a constrained form,
2. the dominant forcing nation controls equilibrium stability.

9 Geopolitical Mechanisms of Quartic Forcing

A fourth strategic nation may force quartic closure through:

1. GPU denial,
2. semiconductor export controls,
3. reserve-currency sanctions,
4. payment-network exclusion,
5. cloud infrastructure denial,
6. AI model dependence,
7. cybernetic surveillance,
8. narrative suppression,
9. financial-information warfare.

These mechanisms alter quartic coefficients dynamically.

10 Graph-Theoretic Interpretation

Let:

$$G = (V, E)$$

represent the global informational graph.

Nodes:

$$V$$

represent nations.

Edges:

$$E$$

represent informational dependencies.

Define narrative centrality:

$$N_i = \frac{\deg(i)}{\sum_j \deg(j)}$$

As informational concentration rises:

1. graph centralization increases,
2. dependency edges concentrate,
3. sovereignty reducibility rises.

11 Financial-Information Dynamics

Modern finance functions as informational infrastructure.

Define financial leverage:

$$F_i = \delta_1 R_i + \delta_2 S_i + \delta_3 P_i + \delta_4 B_i$$

where:

R_i = Reserve currency leverage

S_i = Sanctions capability

P_i = Payment network control

B_i = Banking informational centrality

Reserve-currency systems amplify sovereignty concentration through recursive capital flows.

12 AI/ML and Algorithmic Concentration

AI scaling laws exhibit superlinear concentration effects.

Key drivers include:

1. data accumulation,
2. talent clustering,
3. compute scaling,
4. semiconductor economies of scale,
5. cloud-network effects.

The resulting system resembles computational oligarchy.

13 Warfare Economics

Military-AI integration transforms warfare economics.

Modern conflict increasingly depends upon:

1. autonomous drones,
2. ISR systems,
3. cyberwarfare,

4. satellite intelligence,
5. strategic simulation,
6. AI-guided targeting.

Military capability becomes recursively dependent upon computational sovereignty.

14 International Relations Implications

Traditional international relations theory assumes formal sovereignty equality.

The AI age introduces:

1. computational hierarchy,
2. informational asymmetry,
3. cybernetic dependency,
4. algorithmic imperial structures.

The resulting world order resembles layered informational civilizations.

15 Philosophical Implications

The framework suggests that sovereignty itself becomes computational.

Political independence without informational autonomy becomes structurally incomplete.

Thus:

1. sovereignty becomes recursive,
2. autonomy becomes computational,
3. civilization becomes algorithmic.

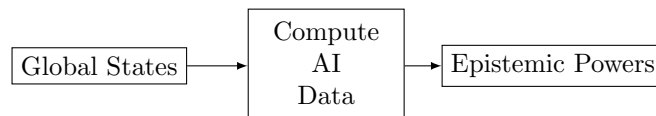
16 Illustrative Comparative Table

Nation	AI	Compute	Data	Narrative	Military	Finance
United States	High	High	High	High	High	High
China	High	High	High	High	High	Medium
Russia	Medium	Medium	Medium	Medium	High	Medium
India	Medium	Medium	High	Medium	Medium	Low
Japan	Medium	High	Medium	Medium	Medium	Medium
South Korea	Medium	High	Medium	Low	Medium	Low

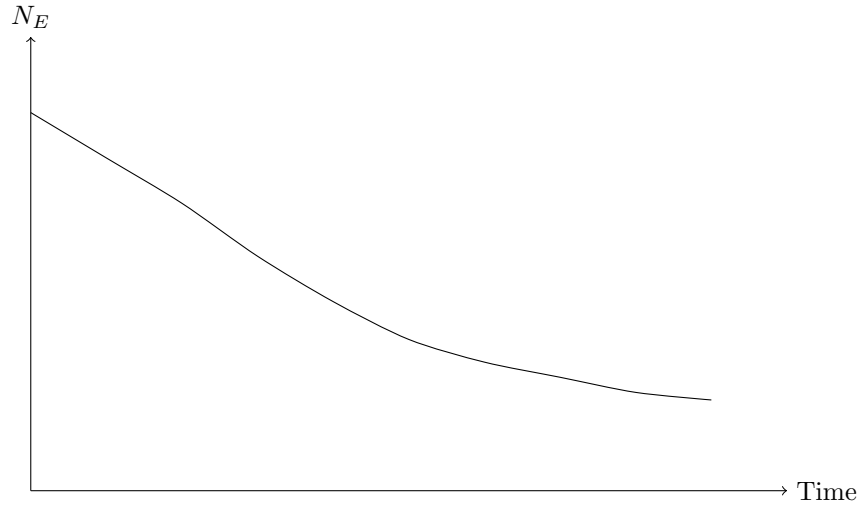
Table 1: Illustrative epistemic sovereignty structure

17 Vector Graphics

17.1 Epistemic Compression Diagram



17.2 Quartic Sovereignty Compression



18 Conclusion

This paper developed a theoretical framework for quartic sovereignty elimination under advanced AI concentration dynamics.

By integrating:

1. AI scaling laws,
2. computational economics,
3. geopolitics,
4. warfare economics,
5. network theory,
6. cybernetics,
7. financial-information systems,
8. international relations,
9. algorithmic concentration,

we formalized the emergence of quartic closure relations in residual sovereignty systems.

The framework predicts:

1. sovereignty compression,
2. informational oligarchy,
3. computational hierarchy,
4. recursive geopolitical dependence,
5. algebraic closure of sovereignty equilibria.

The resulting world order may no longer consist of formally equal sovereign states, but rather layered computational civilizations organized around concentrated informational power.

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Glossary

Epistemic Sovereignty

The capacity of a nation-state to autonomously generate, control, and weaponize strategic information.

Compute Sovereignty

Independent access to advanced computational infrastructure.

Quartic Elimination

Reduction of residual sovereignty dynamics to quartic polynomial closure.

Triple-AI/ML

A framework describing concentrated informational blocs formed through AI and machine learning systems.

Sovereignty Compression

Reduction in the number of informationally autonomous states.

Financial-Information Leverage

Control over informational dimensions of finance and payment systems.

Military-AI Integration

Integration of AI systems into warfare and strategic operations.

Informational Oligarchy

A geopolitical order dominated by concentrated computational civilizations.

Recursive Sovereignty

The concept that sovereignty depends recursively upon informational infrastructure.

Quartic Closure

Polynomial constraint determining admissible sovereignty equilibria.

The End